

Q1 - 24 June - Shift 1

If $\{a_i\}_{i=1}^n$ where n is an even integer, is an arithmetic progression with common difference 1, and $\sum_{i=1}^n a_i = 192$, $\sum_{i=1}^{n/2} a_{2i} = 120$, then n is equal to:

- (A) 48 (B) 96
(C) 92 (D) 104

Space for your notes:

Q2 - 25 June - Shift 1

If $\frac{1}{2 \cdot 3^{10}} + \frac{1}{2^2 \cdot 3^9} + \dots + \frac{1}{2^{10} \cdot 3} = \frac{K}{2^{10} \cdot 3^{10}}$, then the remainder when K is divided by 6 is

- (A) 1 (B) 2
(C) 3 (D) 5

Space for your notes:

Q3 - 25 June - Shift 1

The greatest integer less than or equal to the sum of first 100 terms of the sequence

$\frac{1}{3}, \frac{5}{9}, \frac{19}{27}, \frac{65}{81}, \dots$ is equal to

Space for your notes:

Q4 - 25 June - Shift 2

The sum $1 + 2 \cdot 3 + 3 \cdot 3^2 + \dots + 10 \cdot 3^9$ is equal to

- (A) $\frac{2 \cdot 3^{12} + 10}{4}$ (B) $\frac{19 \cdot 3^{10} + 1}{4}$
(C) $5 \cdot 3^{10} - 2$ (D) $\frac{9 \cdot 3^{10} + 1}{2}$

Space for your notes:

Q5 - 26 June - Shift 1

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Questions

MathonGo

Let $A = \sum_{i=1}^{10} \sum_{j=1}^{10} \min\{i, j\}$ and

$B = \sum_{i=1}^{10} \sum_{j=1}^{10} \max\{i, j\}$. Then $A + B$ is equal to

_____.

Space for your notes:

Q6 - 26 June - Shift 2

If $A = \sum_{n=1}^{\infty} \frac{1}{\left(3 + (-1)^n\right)^n}$ and $B = \sum_{n=1}^{\infty} \frac{(-1)^n}{\left(3 + (-1)^n\right)^n}$, then

$\frac{A}{B}$ is equal to :

(A) $\frac{11}{9}$ (B) 1 (C) $-\frac{11}{9}$ (D) $-\frac{11}{3}$

Space for your notes:

Q7 - 26 June - Shift 2

If $a_1 (> 0)$, a_2, a_3, a_4, a_5 are in a G.P., $a_2 + a_4 = 2a_3 + 1$ and $3a_2 + a_3 = 2a_4$, then $a_2 + a_4 + 2a_5$ is equal to

_____.

Space for your notes:

Q8 - 27 June - Shift 1

Questions

MathonGo

If $x = \sum_{n=0}^{\infty} a^n, y = \sum_{n=0}^{\infty} b^n, z = \sum_{n=0}^{\infty} c^n$, where a, b, c

are in A.P. and $|a| < 1, |b| < 1, |c| < 1, abc \neq 0$, then

(A) x, y, z are in A.P.

(B) x, y, z are in G.P.

(C) $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ are in A.P.

(D) $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1 - (a + b + c)$

Space for your notes:

Q9 - 27 June - Shift 1

If the sum of the first ten terms of the series

$\frac{1}{5} + \frac{2}{65} + \frac{3}{325} + \frac{4}{1025} + \frac{5}{2501} + \dots$ is $\frac{m}{n}$, where

m and n are co-prime numbers, then $m + n$ is equal to _____.

Space for your notes:

Q10 - 27 June - Shift 2

Let $S = 2 + \frac{6}{7} + \frac{12}{7^2} + \frac{20}{7^3} + \frac{30}{7^4} + \dots$ then $4S$ is equal to

(A) $\left(\frac{7}{3}\right)^2$

(B) $\frac{7^3}{3^2}$

(C) $\left(\frac{7}{3}\right)^3$

(D) $\frac{7^2}{3^3}$

Space for your notes:

Q11 - 27 June - Shift 2

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Questions

MathonGo

If $a_1, a_2, a_3, \dots$ and $b_1, b_2, b_3, \dots$ are A.P. and

$a_1 = 2, a_{10} = 3, a_1 b_1 = 1 = a_{10} b_{10}$ then $a_4 b_4$ is equal to

(A) $\frac{35}{27}$

(B) 1

(C) $\frac{27}{28}$

(D) $\frac{28}{27}$

Space for your notes:

Q12 - 28 June - Shift 1

Let $A_1, A_2, A_3, \dots$ be an increasing geometric progression of positive real numbers. If

$A_1 A_3 A_5 A_7 = \frac{1}{1296}$ and $A_2 + A_4 = \frac{7}{36}$, then, the

value of $A_6 + A_8 + A_{10}$ is equal to

(A) 33

(B) 37

(C) 43

(D) 47

Space for your notes:

Q13 - 28 June - Shift 2

If n arithmetic means are inserted between a and 100 such that the ratio of the first mean to the last

mean is $1 : 7$ and $a + n = 33$, then the value of n is

(A) 21

(B) 22

(C) 23

(D) 24

Space for your notes:

Q14 - 28 June - Shift 2

Questions

MathonGo

Let for $n = 1, 2, \dots, 50$, S_n be the sum of the infinite geometric progression whose first term is n^2 and whose common ratio is $\frac{1}{(n+1)^2}$. Then the

Space for your notes:

value of $\frac{1}{26} + \sum_{n=1}^{50} \left(S_n + \frac{2}{n+1} - n - 1 \right)$ is equal to

Q15 - 29 June - Shift 1

Let $\{a_n\}_{n=0}^{\infty}$ be a sequence such that $a_0 = a_1 = 0$ and

Space for your notes:

$a_{n+2} = 2a_{n+1} - a_n + 1$ for all $n \geq 0$. Then, $\sum_{n=2}^{\infty} \frac{a_n}{7^n}$ is

equal to

(A) $\frac{6}{343}$

(B) $\frac{7}{216}$

(C) $\frac{8}{343}$

(D) $\frac{49}{216}$

Q16 - 29 June - Shift 2

The sum of the infinite series

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$1 + \frac{5}{6} + \frac{12}{6^2} + \frac{22}{6^3} + \frac{35}{6^4} + \frac{51}{6^5} + \frac{70}{6^6} + \dots$ is equal to:

(A) $\frac{425}{216}$

(B) $\frac{429}{216}$

(C) $\frac{288}{125}$

(D) $\frac{280}{125}$

Q17 - 29 June - Shift 2

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Questions

MathonGo

Let 3, 6, 9, 12, ... upto 78 terms and 5, 9, 13, 17, ... upto 59 terms be two series. Then, the sum of the terms common to both the series is equal to ____.

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Answer Key

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Q1 (B) Q2 (D) Q3 (98) Q4 (B)
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Q5 (1100) Q6 (C) Q7 (40) Q8 (C)
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Q9 (276) Q10 (C) Q11 (D) Q12 (C)
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Q13 (C) Q14 (41651) Q15 (B) Q16 (C)
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Q17 (2223)
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Q1 (B)

$$\sum_{i=1}^n a_i = \frac{n}{2} \{2a_1 + (n+1)\} = 192$$

$$\Rightarrow 2a_1 + (n-1) = \frac{384}{n} \text{ ----(1)}$$

$$\sum_{i=1}^{n/2} a_{2i} = \frac{n}{4} \left[2a_1 + 2 + \left(\frac{n}{2} - 1 \right) 2 \right] = 120$$

$$2a_1 + n = \frac{480}{n} \text{ ----(2)}$$

From equation (2) and (1)

$$1 = \frac{480}{n} - \frac{384}{n}$$

$$n = 480 - 384 = 96$$

Q2 (D)

$$\frac{1}{2 \cdot 3^{10}} + \frac{1}{2^2 \cdot 3^9} + \frac{1}{2^3 \cdot 3^8} + \dots + \frac{1}{2^{10} \cdot 3} = \frac{K}{2^{10} \cdot 3^{10}}$$

$$K = 2^9 + 2^8 \cdot 3 + 2^7 \cdot 3^2 + \dots + 3^9$$

$$= \frac{2^9 \left(\left(\frac{3}{2} \right)^{10} - 1 \right)}{\frac{3}{2} - 1} = 3^{10} - 2^{10}$$

$$\text{Now, } 3^{10} - 2^{10} = (3^5 - 2^5)(3^5 + 2^5)$$

$$= (211)(275)$$

$$= (35 \times 6 + 1)(45 \times 6 + 5)$$

$$= 6\lambda + 5$$

Remainder is 5.

Q3 (98)

$$\frac{1}{3} + \frac{5}{9} + \frac{19}{27} + \frac{65}{81} + \dots$$

$$\left(1 - \frac{2}{3}\right) + \left(1 - \frac{4}{9}\right) + \left(1 - \frac{8}{27}\right) + \left(1 - \frac{16}{81}\right) \dots 100 \text{ terms}$$

$$100 - \left[\frac{2}{3} + \left(\frac{2}{3}\right)^2 + \dots \right]$$

$$100 - \frac{\frac{2}{3} \left(1 - \left(\frac{2}{3}\right)^{100} \right)}{1 - \frac{2}{3}}$$

$$100 - 2 \left(1 - \left(\frac{2}{3}\right)^{100} \right)$$

$$S = 98 + 2 \left(\frac{2}{3}\right)^{100}$$

$$\Rightarrow [S] = 98$$

Q4 (B)

$$S = 1 \cdot 3^0 + 2 \cdot 3^1 + 3 \cdot 3^2 + \dots + 10 \cdot 3^9$$

$$3S = 1 \cdot 3^1 + 2 \cdot 3^2 + \dots + 9 \cdot 3^9 + 10 \cdot 3^{10}$$

$$-2S = (1 \cdot 3^0 + 3^1 + 3^2 + \dots + 3^9) - 10 \cdot 3^{10}$$

$$S = 5 \times 3^{10} - \left(\frac{3^{10} - 1}{4} \right)$$

$$S = \frac{20 \cdot 3^{10} - 3^{10} + 1}{4} = \frac{19 \cdot 3^{10} + 1}{4}$$

Q5 (1100)

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Hints and Solutions

MathonGo

$$A = \sum_{i=1}^{10} \sum_{j=1}^{10} \min\{i, j\}$$

$$B = \sum_{i=1}^{10} \sum_{j=1}^{10} \max\{i, j\}$$

$$A = \sum_{j=1}^{10} \min(i, 1) + \min(j, 2) + \dots \min(i, 10)$$

$$= \underbrace{(1+1+1+\dots+1)}_{19 \text{ times}} + \underbrace{(2+2+2+\dots+2)}_{17 \text{ times}} + \underbrace{(3+3+3+\dots+3)}_{15 \text{ times}}$$

$$+ \dots (1) \text{ 1 times}$$

$$B = \sum_{j=1}^{10} \max(i, 1) + \max(j, 2) + \dots \max(i, 10)$$

$$= \underbrace{(10+10+\dots+10)}_{19 \text{ times}} + \underbrace{(9+9+\dots+9)}_{17 \text{ times}} + \dots + 1 \text{ 1 times}$$

$$A + B = 20(1 + 2 + 3 + \dots + 10)$$

$$= 20 \times \frac{10 \times 11}{2} = 10 \times 110 = 1100$$

Q6 (C)

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$$A = \left(\frac{1}{2} + \frac{1}{4^2} + \frac{1}{2^3} + \frac{1}{4^4} + \dots \infty \right)$$

$$A = \left(\frac{1}{2} + \frac{1}{2^3} + \dots \infty \right) + \left(\frac{1}{4^2} + \frac{1}{4^4} + \dots \infty \right)$$

$$A = \left(\frac{\frac{1}{2}}{1 - \frac{1}{4}} + \frac{\frac{1}{16}}{1 - \frac{1}{16}} \right)$$

$$\Rightarrow A = \frac{1}{2} \times \frac{4}{3} + \frac{1}{16} \times \frac{16}{15} \Rightarrow A = \frac{11}{15}$$

$$B = \frac{-1}{2} + \frac{1}{4^2} + \frac{-1}{2^3} + \frac{1}{4^4} + \dots \infty$$

$$B = \left(\frac{-1}{2} + \frac{-1}{2^3} + \dots \infty \right) + \left(\frac{1}{4^2} + \frac{1}{4^4} + \dots \infty \right)$$

$$B = \frac{-\frac{1}{2}}{1 - \frac{1}{4}} + \frac{\frac{1}{16}}{1 - \frac{1}{16}}$$

$$\Rightarrow B = -\frac{1}{2} \times \frac{4}{3} + \frac{1}{16} \times \frac{16}{15}$$

$$B = -\frac{9}{15}$$

$$\frac{A}{B} = \frac{11}{15} \times \frac{15}{(-9)}$$

$$\frac{A}{B} = -\frac{11}{9}$$

Q7 (40)

$$a_1 > 0, a_2, a_3, a_4, a_5 \rightarrow \text{G.P.}$$

$$3a_2 + a_3 = 2a_4$$

$$3ar + ar^2 = 2ar^3$$

$$3 + r = 2r^2$$

$$2r^2 - r - 3 = 0$$

$$r = -1 \text{ \& } r = \frac{3}{2}$$

$$a_2 + a_4 = 2a_3 + 1$$

$$ar + ar^3 = 2ar^2 + 1$$

$$a(r + r^3 - 2r^2) = 1$$

$$a\left(\frac{3}{2} + \frac{27}{8} - \frac{18}{4}\right) = 1$$

$$a = \frac{8}{3}$$

$$\text{When } r = -1, a = -\frac{1}{4} \text{ (rejected, } a_1 > 0)$$

$$r = \frac{2}{3}, a = \frac{8}{3} \text{ (selected)}$$

Now

$$a_2 + a_4 + 2a_5$$

$$= \frac{8}{3} \times \frac{2}{3} + \frac{8}{3} \times \frac{27}{8} + 2 \times \frac{8}{3} \times \frac{81}{16}$$

$$= 4 + 9 + 27 = 40$$

Q8 (C)

$$x = 1 + a + a^2 = \dots\dots\dots$$

$$x = \frac{1}{1-a} \Rightarrow a = 1 - \frac{1}{x}$$

$$y = \frac{1}{1-b} \Rightarrow b = 1 - \frac{1}{y}$$

$$z = \frac{1}{1-c} \Rightarrow c = 1 - \frac{1}{z}$$

a, b, c are in A.P.

$$\Rightarrow 1 - \frac{1}{x}, 1 - \frac{1}{y}, 1 - \frac{1}{z} \text{ are in A.P.}$$

$$\Rightarrow -\frac{1}{x}, -\frac{1}{y}, -\frac{1}{z} \text{ are in A.P.}$$

$$\Rightarrow \frac{1}{x}, \frac{1}{y}, \frac{1}{z} \text{ are in A.P.}$$

Q9 (276)

$$\frac{1}{5} + \frac{2}{65} + \frac{3}{325} + \frac{4}{1025} + \frac{5}{2501} + \dots$$

$$T_n = \frac{n}{4n^4 + 1}$$

$$= \frac{n}{(2n^2 + 1)^2 - (2n)^2} = \frac{n}{(2n^2 + 2n + 1)(2n^2 - 2n + 1)}$$

$$= \frac{1}{4} \left[\frac{1}{2n^2 - 2n + 1} - \frac{1}{2n^2 + 2n + 1} \right]$$

$$S_{10} = \sum_{n=1}^{10} T_n = \frac{1}{4} \left[\frac{1}{1} - \frac{1}{5} + \frac{1}{5} - \frac{1}{13} + \dots - \frac{1}{200 + 20 + 1} \right]$$

$$= \frac{1}{4} \left[1 - \frac{1}{221} \right] = \frac{1}{4} \times \frac{220}{221} = \frac{55}{221} = \frac{m}{n}$$

$$m + n = 55 + 221 = 276$$

Q10 (C)

$$S = 2 + \frac{6}{7} + \frac{12}{7^2} + \frac{20}{7^3} + \frac{30}{7^4} + \dots$$

Considering infinite sequence,

$$S = 2 + \frac{6}{7} + \frac{12}{7^2} + \frac{20}{7^3} + \frac{30}{7^4} + \dots$$

$$\frac{S}{7} = \frac{2}{7} + \frac{6}{7^2} + \frac{12}{7^3} + \frac{20}{7^4} + \dots$$

$$\Rightarrow \frac{6S}{7} = 2 + \frac{4}{7} + \frac{6}{7^2} + \frac{8}{7^3} + \frac{10}{7^4} + \dots$$

$$\Rightarrow \frac{6S}{7^2} = \frac{2}{7} + \frac{4}{7^2} + \frac{6}{7^3} + \frac{8}{7^4} + \dots$$

$$\frac{6S}{7} \left(1 - \frac{1}{7} \right) = 2 + \frac{2}{7} + \frac{2}{7^2} + \frac{2}{7^3} + \dots$$

$$\Rightarrow \frac{6^2 S}{7^2} = \frac{2}{1 - \frac{1}{7}} = \frac{2}{6} \times 7$$

$$\Rightarrow S = \frac{2 \times 7^3}{6^3} \Rightarrow 4S = \frac{7^3}{3^3} = \left(\frac{7}{3} \right)^3$$

Q11 (D)

Hints and Solutions

MathonGo

$$a_1, a_2, a_3 \dots \text{A.P.}; a_1 = 2; a_{10} = 3; d_1 = \frac{1}{9}$$

$$b_1, b_2, b_3, \dots, \text{A.P.}; b_1 = \frac{1}{2}; b_{10} = \frac{1}{3}; d_2 = \frac{-1}{54}$$

[Using $a_1 b_1 = 1 = a_{10} b_{10}$; d_1 & d_2 are common differences respectively]

$$a_4 \cdot b_4 = (2 + 3d_1) \left(\frac{1}{2} + 3d_2 \right)$$

$$= \left(2 + \frac{1}{3} \right) \left(\frac{1}{2} - \frac{1}{18} \right)$$

$$= \left(\frac{7}{3} \right) \left(\frac{8}{18} \right) = \frac{28}{27}$$

Q12 (C)

$$A_1 \cdot A_3 \cdot A_5 \cdot A_7 = \frac{1}{1296}$$

$$(A_4)^4 = \frac{1}{1296}$$

$$A_4 = \frac{1}{6} \dots (1)$$

$$A_2 + A_4 = \frac{7}{36}$$

$$A_2 = \frac{1}{36} \dots (2)$$

$$A_6 = 1$$

$$A_8 = 6$$

$$A_{10} = 36$$

$$A_6 + A_8 + A_{10} = 43$$

Q13 (C)

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$$d = \frac{100 - a}{n + 1}$$

$$A_1 = a + d$$

$$A_n = 100 - d$$

$$\Rightarrow \frac{A_1}{A_n} = \frac{1}{7} \Rightarrow \frac{a + d}{100 - d} = \frac{1}{7}$$

$$\Rightarrow 7a + 8d = 100$$

$$\Rightarrow 7a + 8\left(\frac{100 - a}{n + 1}\right) = 100 \quad \dots(1)$$

$$\because a + n = 33 \quad \dots(2)$$

Now, by Eq. (1) and (2)

$$7n^2 - 132n - 667 = 0$$

$$\boxed{n = 23} \text{ and } n = \frac{-29}{7} \text{ reject.}$$

Q14 (41651)

$$S_n = \frac{n^2}{1 - \frac{1}{(n+1)^2}} = \frac{n(n+1)^2}{(n+2)}$$

$$S_n = \frac{n(n^2 + 2n + 1)}{(n+2)}$$

$$S_n = \frac{n[n(n+2) + 1]}{(n+2)}$$

$$S_n = n \left[n + \frac{1}{n+2} \right]$$

$$S_n = n^2 + \frac{n+2-2}{(n+2)}$$

$$S_n = n^2 + 1 - \frac{2}{(n+2)}$$

$$\text{Now } \frac{1}{26} + \sum_{n=1}^{50} \left[(n^2 - n) - 2 \left(\frac{1}{n+2} - \frac{1}{n+1} \right) \right]$$

$$= \frac{1}{26} + \left[\frac{50 \times 51 \times 101}{6} - \frac{50 \times 51}{2} - 2 \left(\frac{1}{52} - \frac{1}{2} \right) \right]$$

$$= 41651$$

Q15 (B)

Hints and Solutions

MathonGo

$$a_2 = 1, a_3 = 3, a_4 = 6$$

$$a_n = \frac{n(n-1)}{2}$$

$$S = \sum_{n=2}^{\infty} \frac{n(n-1)}{2(7^n)}$$

$$S = \frac{1}{7^2} + \frac{3}{7^3} + \frac{6}{7^4} + \frac{10}{7^5} + \frac{15}{7^6} + \dots$$

$$\frac{S}{7} = \frac{1}{7^3} + \frac{3}{7^4} + \frac{6}{7^5} + \frac{10}{7^6} + \dots$$

$$6\frac{S}{7} = \frac{1}{7^2} + \frac{2}{7^3} + \frac{3}{7^4} + \frac{4}{7^5} + \dots$$

$$6\frac{S}{7^2} = \frac{1}{7^3} + \frac{2}{7^4} + \frac{3}{7^5} + \dots$$

$$6\frac{S}{7} \cdot \frac{6}{7} = \frac{1}{7^2} + \frac{1}{7^3} + \dots = \frac{1/7^2}{1-1/7}$$

$$6 \times 6 \frac{S}{7^2} = \frac{1}{7 \times 6}$$

$$S = \frac{7}{6^3} = \frac{7}{216}$$

Alternate

$$a_{n+2} = 2a_{n+1} - a_n + 1$$

$$\Rightarrow \frac{a_{n+2}}{7^{n+2}} = \frac{2}{7} \frac{a_{n+1}}{7^{n+1}} - \frac{1}{49} \frac{a_n}{7^n} + \frac{1}{7^{n+2}}$$

$$\Rightarrow \sum_{n=2}^{\infty} \frac{a_{n+2}}{7^{n+2}} = \frac{2}{7} \sum_{n=2}^{\infty} \frac{a_{n+1}}{7^{n+1}} - \frac{1}{49} \sum_{n=2}^{\infty} \frac{a_n}{7^n} + \sum_{n=2}^{\infty} \frac{1}{7^{n+2}}$$

$$\text{Let } \sum_{n=2}^{\infty} \frac{a_n}{7^n} = p$$

$$\Rightarrow \left(p - \frac{a_2}{7^2} - \frac{a_3}{7^3} \right) = \frac{2}{7} \left(p - \frac{a_2}{7^2} \right) - \frac{1}{49} p + \frac{1/7^4}{1 - \frac{1}{7}}$$

$$\therefore a_2 = 1, a_3 = 3$$

$$\Rightarrow p - \frac{1}{49} - \frac{3}{7} = \frac{2}{7} p - \frac{2}{7^3} - \frac{p}{49} + \frac{1}{7^3}$$

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Q16 (C)

$$S = 1 + \frac{5}{6} + \frac{12}{6^2} + \frac{22}{6^3} + \frac{35}{6^4} + \dots$$

$$\frac{S}{6} = \frac{1}{6} + \frac{5}{6^2} + \frac{12}{6^3} + \frac{22}{6^4} + \dots$$

on subtraction

$$\frac{5}{6}S = 1 + \frac{4}{6} + \frac{7}{6^2} + \frac{10}{6^3} + \frac{13}{6^4} + \dots$$

$$\frac{5}{36}S = 1 + \frac{4}{6^2} + \frac{7}{6^3} + \frac{10}{6^4} + \frac{13}{6^5} + \dots$$

on subtraction

$$\frac{25}{36}S = 1 + \frac{3}{6} + \frac{3}{6^2} + \frac{3}{6^3} + \dots = \frac{8}{5}$$

$$S = \frac{288}{125}$$

Q17 (2223)

For series of common terms

$$a=9, d=12, n=19$$

$$S_{19} = \frac{19}{2}[2(9) + 18(12)] = 2223$$